

Aarab Nishchal

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Technical Skills

Languages: JavaScript, TypeScript, Python, SQL, HTML, CSS
Web & Frameworks: React, Next.js, Node.js, Express, Angular, Django, FastAPI, Tailwind CSS
Databases & Cloud: PostgreSQL, MongoDB, Supabase, ChromaDB, Convex, AWS EC2
AI / ML & LLMs: PyTorch, LangChain, Ollama, Groq, VAPI, Whisper, Gemini, OpenRouter, RNN/LSTM
Tools: Git, GitHub, CI/CD, Jenkins, n8n, REST APIs, Postman, Jira

Experience

Unstop Sep 2024 – Present
Software Developer Intern Remote

- Integrated AI-driven **n8n** automation workflows across internal REST API endpoints and a custom CMS, adding white-label support and dynamic window-title configuration
- Built an AI workflow system to auto-generate and validate vendor assignments, cutting creation and validation time by **60 to 70 %**
- Enhanced front-end UI components to surface the new workflow system, streamlining vendor-facing task flows
- Managed CI/CD pipelines via Git and Jenkins, maintaining consistent deployments across environments
- Tracked sprint progress and maintained technical documentation in Jira and Confluence

Projects

TestIQ | *LangChain · ChromaDB · Ollama · Tree-sitter · Python* GitHub

- Built a CLI **RAG pipeline** using **LangChain** and **ChromaDB** to analyze codebases and auto-generate unit tests, achieving 60 passing tests across 4 development phases
- Engineered a self-correcting validation loop with **Tree-sitter** (AST parsing) and **Ollama** (local inference) to eliminate invalid test output before finalization

VidyaMarg | *Next.js · FastAPI · Groq (LLaMA 3.1 70B) · NetworkX · Supabase · React Flow* GitHub

- Developed a full-stack AI learning path generator integrating **LLaMA 3.1 70B** via **Groq**, producing structured goal-based roadmaps from user input
- Architected graph-based path rendering with **NetworkX** and **React Flow**, visualizing learning paths as interactive **DAGs** backed by **Supabase**

Curely | *VAPI · OpenAI Whisper · Gemini · Openrouter* GitHub

- Designed an end-to-end real-time voice AI assistant for medical Q&A using **VAPI** and **Whisper**, with **Gemini** handling response generation across the full voice pipeline
- Reduced inference cost to **\$0.08/min** through targeted model selection and request routing via **Openrouter**

Ideascribe | *Convex · Clerk · React · Blocknote · Edgestore* GitHub

- Engineered a note taking application for developers, integrating **BlockNote** for dynamic rich-text editing, **Convex** for rapid state synchronization, and **Edge Store** for scalable cloud asset management.

Education

B.Tech, Computer Science & Engineering 2022 – 2026
Kalinga Institute of Industrial Technology
Senior Secondary (Class XII) 2020 – 2021
BSEB, BMP-7, Katihar, Bihar
Secondary (Class X) 2019 – 2020
CBSE, New Light Academy, Namchi, Sikkim

Positions of Responsibility

Team Lead – *College Project Team, KIIT* 2023 – 2024

- Led a 6-member team across 3 college projects, managing task allocation, code reviews, and delivery milestones

Lead Graphic Designer & Social Media Manager – *Fashion Society, KIIT* 2022 – 2024

- Directed visual identity and social media strategy for college fashion events across two semesters

Volunteer – *Youth Red Cross Society, KIIT* 2022 – 2023

- Led a city-wide cloth donation drive, coordinating outreach and logistics across multiple residential zones

Certifications

- AWS Certified Cloud Practitioner** – Amazon Web Services
- Scientific Computing with Python** – freeCodeCamp
- AMCAT Certified** in Software Development & Database Management – Aspiring Minds
- Problem Solving Certificate** – HackerRank
- Keplay Fellowship Program** – Open-source API testing and developer advocacy